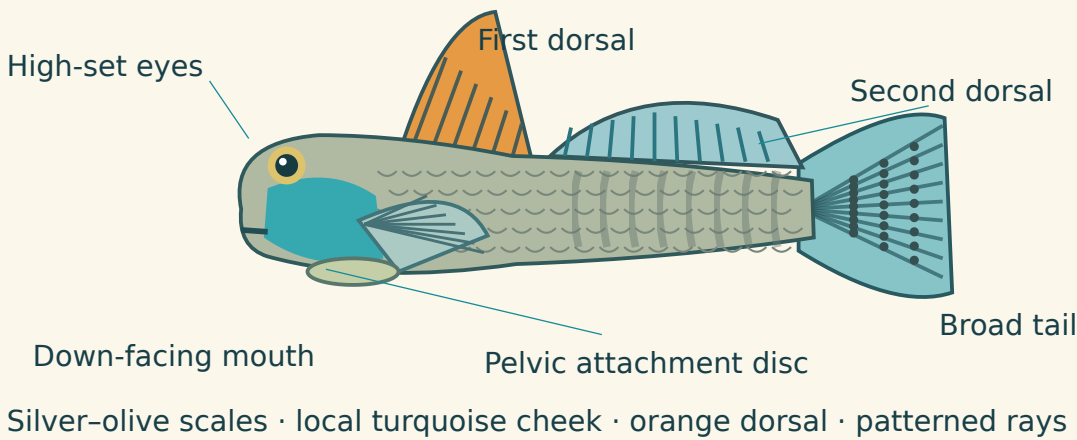


Rainbow goby

Stiphodon ornatus · A small fish with a very different way of staying put.

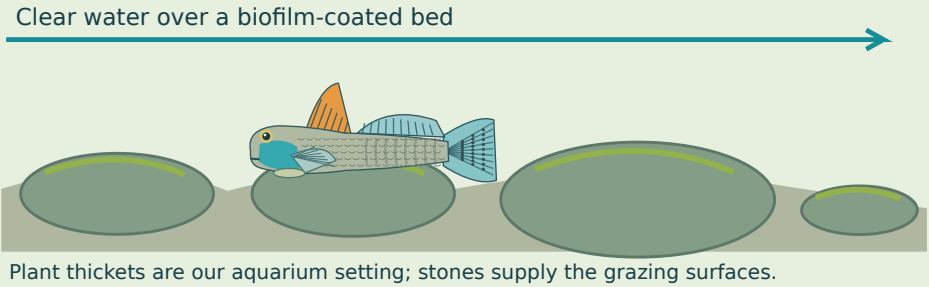
ADULT MALE · STYLIZED COLOUR STUDY



A name needs a species

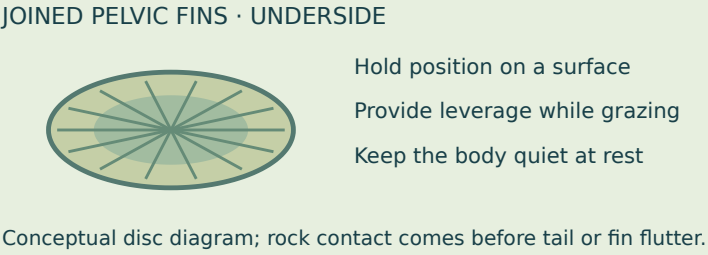
“Rainbow goby” covers several traded fish. Our reference is an adult male *S. ornatus*, documented from western Sumatra. [1,2]
Examined males were about 35–53 mm in standard length, excluding the tail. The male tail is roughly 29–35% of that length. [1]
Original drawings. The chosen turquoise cheek and warm dorsal are a live-colour interpretation, not a diagnostic colour chart. [3]

01 · A rock is a feeding station



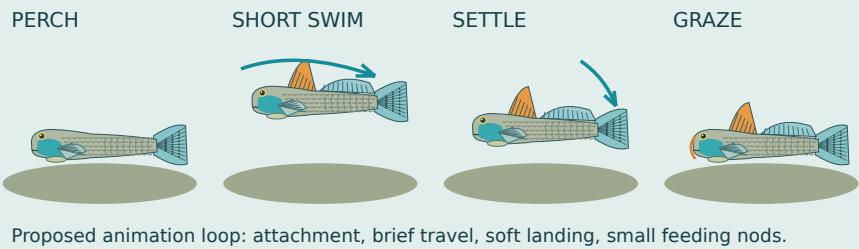
Clear, oxygen-rich streams support algae and associated microscopic life on submerged surfaces. The fish grazes this biofilm. [2]
For our planted aquarium, leave rounded stones visible among the colonies. This is an aquarium interpretation of its habitat.

02 · Attach, then graze



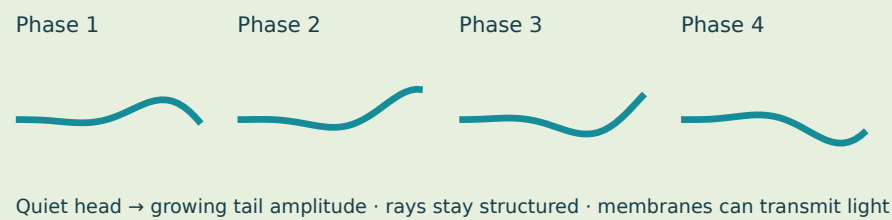
The pelvic fins form a disc. It helps the fish hold position and brace during feeding; its mouth faces the substrate. [1,2]
A convincing resting pose needs a believable contact point. Tiny feeding motions should leave the whole fish settled.

03 · Motion with pauses



The grazing habit motivates this game loop: perch → brief swim → settle → graze. Timing, acceleration and descent rates are authored choices. Avoid perpetual cruising, instant direction flips and sliding while apparently attached.

04 · Structure before shimmer



Use two distinct dorsals, a low elongated body, broad tail and patterned fin rays. [1] Keep body scales restrained; concentrate bright colour around the cheek and dorsal.
Game proposal: 256² authored textures, selective fin translucency, bounded tail deformation and modest pectoral motion.

05 · From biology to controls

← → · Short lateral swims, easing into movement.
↑ ↓ · Lift off or descend toward a perch.
Release · Settle; a quiet graze follows contact.
Tap A · Change view. **Hold A** · Settings.
↻ · Close settings first; otherwise return to the level selector.
Proposed controls preserve the plant level’s camera/settings gestures. Forth reads the Director’s OAS/OAD Water type: freshwater chooses the goby; saltwater chooses the urchin.

06 · Small Forth ideas: ease, settle, keep the head quiet

```
\ Mailbox velocity eases toward intent
: gb-ease ( target mailbox -- )
>r r@ gb@ -
gb-dt gb@ 7 * 1 min *
r@ gb@ + r> gb! ;

\ Illustrative tail envelope: u in [0,1]
: tail-weight ( u -- amplitude ) dup * ;
```

The first fragment comes from the authored controller. The second is a simple deformation idea: little motion near the head, more toward the tail. Neither is a measured biological law.